TaskFlow — Entities, Attributes and Relationships

Part 1: each entity with its attributes. Part 2: all relationships with their type (one-to-one, one-to-many, many-to-many) and participation (total or partial) for each side.

Part 1: Entities and Attributes

1. User

- User Identifier (Primary Key)
- Full Name (Composite)
- Email Address (Candidate Key)
- Password Credential
- Theme Preference
- Account Status
- Registration Date
- Current Workload (Derived)

2. Role

- Role Identifier (Primary Key)
- Role Name (Candidate Key)
- Hierarchy Level
- Role Description

3. Permission

- Permission Identifier (Primary Key)
- Permission Name (Candidate Key)
- Permission Description

4. Workspace

- Workspace Identifier (Primary Key)
- Workspace Name
- Workspace Description
- Creation Date
- Workspace Settings (Composite)

5. Team

- Team Identifier (Primary Key)
- Team Name
- Team Description
- Creation Date
- Member Count (Derived)

6. Team Membership

- Membership Identity (Primary Key)
- Join Date
- Membership Status

- Last Activity Date (Derived)

7. Invitation

- Invitation Identifier (Primary Key)
- Invitee Email Address
- Sent Date
- Expiry Date
- Invitation Status

8. Project

- Project Identifier (Primary Key)
- Project Name
- Project Description
- Planned Start Date
- Planned End Date
- Project Settings (Composite)
- Archived Status
- Archived Date
- Creation Date
- Progress Percentage (Derived)

9. Project Template

- Project Template Identifier (Primary Key)
- Template Name (Candidate Key)
- Template Description
- Default Column Names (Multivalued)

10. Board

- Board Identifier (Primary Key)
- Board Name
- Board Description
- Display Order
- Creation Date

11. Board Column

- Column Name (Partial Key)
- Column Position
- WIP Limit
- Completion Column Flag
- Current Task Count (Derived)

12. Swimlane

- Swimlane Name (Partial Key)
- Swimlane Position
- Categorization Criterion

13. Task

- Task Identifier (Primary Key)
- Title
- Description
- Priority
- Status (Derived)
- Due Date
- Start Date
- Estimated Time
- Actual Time Spent (Derived)
- Duration (Derived)
- Checklist Progress (Derived)
- Overdue Indicator (Derived)
- Board Position
- Archived Status
- Archived Date
- Creation Date
- Completion Date
- Last Modified Date
- Row Version

14. Checklist Item

- Item Title (Partial Key)
- Completion Status
- Item Position
- Completed Date

15. Label

- Label Identifier (Primary Key)
- Label Name
- Label Color
- Label Description

16. Custom Field

- Custom Field Identifier (Primary Key)
- Field Name
- Field Data Type
- Allowed Values (Multivalued)

17. Task Template

- Task Template Identifier (Primary Key)
- Template Name
- Default Title
- Default Description
- Default Priority
- Default Estimated Time
- Default Checklist Items (Multivalued)
- Recurrence Pattern (Composite)

18. Sprint

- Sprint Identifier (Primary Key)
- Sprint Name
- Sprint Start Date
- Sprint End Date
- Sprint Goal
- Velocity (Derived)

19. Milestone

- Milestone Identifier (Primary Key)
- Milestone Name
- Milestone Date
- Milestone Description
- Milestone Status

20. Comment

- Comment Identifier (Primary Key)
- Comment Content
- Posted Date and Time
- Last Edited Date and Time

21. Attachment

- Attachment Identifier (Primary Key)
- File Details (Composite)
- Stored File Content
- Upload Date and Time

22. Time Entry

- Time Entry Identifier (Primary Key)
- Start Date and Time
- End Date and Time
- Logged Duration
- Entry Method
- Work Note

23. Notification

- Notification Identifier (Primary Key)
- Notification Type
- Message Text
- Created Date and Time
- Read Status
- Delivery Channels (Multivalued)

24. Notification Preference

- Notification Preference Identifier (Primary Key)
- Preference Notification Type
- In-App Enabled

- Email Enabled
- Digest Frequency
- Reminder Lead Time

25. Activity Log Entry

- Activity Log Entry Identifier (Primary Key)
- Action Type
- Action Timestamp
- Changed Data (Composite)
- Action Description

26. Saved Filter

- Saved Filter Identifier (Primary Key)
- Filter Name
- Filter Criteria (Composite)
- Sharing Scope

27. Automation Rule

- Automation Rule Identifier (Primary Key)
- Rule Name
- Trigger Type
- Action Type
- Due Date Lead Time
- Enabled Status
- Creation Date

28. Webhook

- Webhook Identifier (Primary Key)
- Webhook Name
- Target URL
- Subscribed Events (Multivalued)
- Active Status

29. Dashboard Widget

- Dashboard Widget Identifier (Primary Key)
- Widget Type
- Display Position
- Visibility Status

30. System Setting

- System Setting Identifier (Primary Key)
- Setting Name (Candidate Key)
- Setting Value
- Setting Description
- Last Modified Date

- Entity Name
- Entity Id

Part 2: Relationships

1. Owns: User — Workspace

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Workspace: Total participation (1,1)

2. Contains Team: Workspace — Team

Type: One-to-Many (1:N)

Workspace: Partial participation (0,N)

Team: Total participation (1,1)

3. Has Membership: Team — Team Membership

Type: One-to-Many (1:N)

Team: Partial participation (0,N)

Team Membership: Total participation (1,1)

4. Joins As Member: User — Team Membership

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Team Membership: Total participation (1,1)

5. Holds Role: Role — Team Membership

Type: One-to-Many (1:N)

Role: Partial participation (0,N)

Team Membership: Total participation (1,1)

6. Grants Permission: Role — Permission

Type: Many-to-Many (M:N)

Role: Total participation (1,N)

Permission: Partial participation (0,N)

7. Sends Invitation: User — Invitation

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Invitation: Total participation (1,1)

8. Invites To Team: Team — Invitation

Type: One-to-Many (1:N)

Team: Partial participation (0,N)

Invitation: Total participation (1,1)

9. Offers Role: Role — Invitation

Type: One-to-Many (1:N)

Role: Partial participation (0,N)

Invitation: Total participation (1,1)

11. Has Access To: Team — Project

Type: Many-to-Many (M:N)

Team: Partial participation (0,N)

Project: total participation (1,N)

12. Creates Project: User — Project

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Project: Total participation (1,1)

13. Contains Project: Workspace — Project

Type: One-to-Many (1:N)

Workspace: Partial participation (0,N)

Project: Total participation (1,1)

14. Created From Template: Project Template — Project

Type: One-to-Many (1:N)

Project Template: Partial participation (0,N)

Project: Partial participation (0,1)

15. Contains Board: Project — Board

Type: One-to-Many (1:N)

Project: Partial participation (0,N)

Board: Total participation (1,1)

16. Contains Column: Board — Board Column

Type: One-to-Many (1:N)

Board: Partial participation (0,N)

Board Column: Total participation (1,1)

17. Contains Swimlane: Board — Swimlane

Type: One-to-Many (1:N)

Board: Partial participation (0,N)

Swimlane: Total participation (1,1)

18. Contains Sprint: Project — Sprint

Type: One-to-Many (1:N)

Project: Partial participation (0,N)

Sprint: Total participation (1,1)

19. Contains Milestone: Project — Milestone

Type: One-to-Many (1:N)

Project: Partial participation (0,N)

Milestone: Total participation (1,1)

20. Defines Label: Project — Label

Type: One-to-Many (1:N)

Project: Partial participation (0,N)

Label: Total participation (1,1)

21. Defines Custom Field: Project — Custom Field

Type: One-to-Many (1:N)

Project: Partial participation (0,N)

Custom Field: Total participation (1,1)

22. Owns Task Template: Project — Task Template

Type: One-to-Many (1:N)

Project: Partial participation (0,N)

Task Template: Total participation (1,1)

23. Contains Task: Board Column — Task

Type: One-to-Many (1:N)

Board Column: Partial participation (0,N)

Task: Total participation (1,1)

24. Groups Task: Swimlane — Task

Type: One-to-Many (1:N)

Swimlane: Partial participation (0,N)

Task: Partial participation (0,1)

25. Creates Task: User — Task

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Task: Total participation (1,1)

26. Assigned To: Task — User

Type: Many-to-Many (M:N)

Task: Partial participation (0,N)

User: Partial participation (0,N)

Relationship attributes: Assigned Date

27. Is Planned In: Sprint — Task

Type: One-to-Many (1:N)

Sprint: Partial participation (0,N)

Task: Partial participation (0,1)

28. Tagged With: Task — Label

Type: Many-to-Many (M:N)

Task: Partial participation (0,N)

Label: Partial participation (0,N)

29. Has Checklist Item: Task — Checklist Item

Type: One-to-Many (1:N)

Task: Partial participation (0,N)

Checklist Item: Total participation (1,1)

30. Depends On: Task — Task

Type: Many-to-Many (M:N)

Task (Successor): Partial participation (0,N)

Task (Predecessor): Partial participation (0,N)

Relationship attributes: Dependency Type, Lag Time

31. Has Custom Field Value: Task — Custom Field

Type: Many-to-Many (M:N)

Task: Partial participation (0,N)

Custom Field: Partial participation (0,N)

Relationship attributes: Field Value

32. Marks: Milestone — Task

Type: One-to-Many (1:N)

Milestone: Partial participation (0,N)

Task: Partial participation (0,1)

33. Instantiated From: Task Template — Task

Type: One-to-Many (1:N)

Task Template: Partial participation (0,N)

Task: Partial participation (0,1)

34. Has Comment: Task — Comment

Type: One-to-Many (1:N)

Task: Partial participation (0,N)

Comment: Total participation (1,1)

35. Authors: User — Comment

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Comment: Total participation (1,1)

36. Replies To: Comment — Comment

Type: One-to-Many (1:N)

Comment (Parent comment): Partial participation (0,N)

Comment (Reply): Partial participation (0,1)

37. Mentions: Comment — User

Type: Many-to-Many (M:N)

Comment: Partial participation (0,N)

User: Partial participation (0,N)

38. Has Attachment: Task — Attachment

Type: One-to-Many (1:N)

Task: Partial participation (0,N)

Attachment: Partial participation (0,1)

39. Comment Has Attachment: Comment — Attachment

Type: One-to-Many (1:N)

Comment: Partial participation (0,N)

Attachment: Partial participation (0,1)

40. Uploads: User — Attachment

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Attachment: Total participation (1,1)

41. Is Logged On: Task — Time Entry

Type: One-to-Many (1:N)

Task: Partial participation (0,N)

Time Entry: Total participation (1,1)

42. Logs Time: User — Time Entry

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Time Entry: Total participation (1,1)

43. Receives: User — Notification

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Notification: Total participation (1,1)

44. Concerns Task: Task — Notification

Type: One-to-Many (1:N)

Task: Partial participation (0,N)

Notification: Partial participation (0,1)

45. Is Triggered By Comment: Comment — Notification

Type: One-to-Many (1:N)

Comment: Partial participation (0,N)

Notification: Partial participation (0,1)

46. Configures: User — Notification Preference

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Notification Preference: Total participation (1,1)

47. Performs: User — Activity Log Entry

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Activity Log Entry: Partial participation (0,1)

48. Records Activity On Task: Task — Activity Log Entry

Type: One-to-Many (1:N)

Task: Partial participation (0,N)

Activity Log Entry: Partial participation (0,1)

49. Records Activity On Project: Project — Activity Log Entry

Type: One-to-Many (1:N)

Project: Partial participation (0,N)

Activity Log Entry: Partial participation (0,1)

50. Saves Filter: User — Saved Filter

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Saved Filter: Total participation (1,1)

51. Scopes Filter To Project: Project — Saved Filter

Type: One-to-Many (1:N)

Project: Partial participation (0,N)

Saved Filter: Partial participation (0,1)

52. Contains Automation Rule: Project — Automation Rule

Type: One-to-Many (1:N)

Project: Partial participation (0,N)

Automation Rule: Total participation (1,1)

53. Calls Webhook: Webhook — Automation Rule

Type: One-to-Many (1:N)

Webhook: Partial participation (0,N)

Automation Rule: Partial participation (0,1)

54. Configures Webhook: Project — Webhook

Type: One-to-Many (1:N)

Project: Partial participation (0,N)

Webhook: Total participation (1,1)

55. Filters By Label: Automation Rule — Label

Type: Many-to-Many (M:N)

Automation Rule: Partial participation (0,N)

Label: Partial participation (0,N)

56. Targets Column: Automation Rule — Board Column

Type: Many-to-Many (M:N)

Automation Rule: Partial participation (0,N)

Board Column: Partial participation (0,N)

57. Auto-Assigns To: Automation Rule — User

Type: One-to-Many (1:N)

Automation Rule: Partial participation (0,1)

User: Partial participation (0,N)

58. Owns Widget: User — Dashboard Widget

Type: One-to-Many (1:N)

User: Partial participation (0,N)

Dashboard Widget: Total participation (1,1)